

[CO-1]

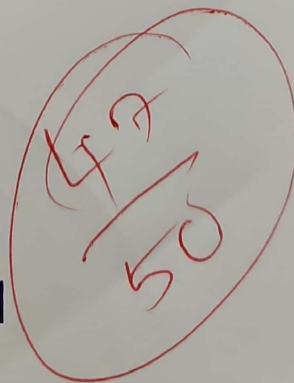
Assessment Tool-01

Name - M. Samanvi

Rg. No - 192425318

slot - B

code - MLA0401



94%
marks

course Name - Deep Learning

1. A coin is tossed 10 times & the outcome are
HHTHHHTHHT

Definition -

MLE is a statistical method used to estimate the value of unknown parameters from data.

Formula -

$$\hat{p} = \frac{\text{Number of successes}}{\text{Total Number of Trials}}$$

Total Tosses = 10

Number of Heads = 7

calculation -

$$\begin{aligned}\hat{p} &= \frac{7}{10} \\ &= 0.7\end{aligned}$$

Answer

$$\boxed{P = 0.7}$$

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Answer

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2. out of 50 student, 42 Passed the exam
MLE estimates the probability of success by
dividing the number of successful

Formula -

$$\hat{p} = \frac{\text{Number of successes}}{\text{Total Observations}}$$

Given,

Total student = 50

Passed student = 42

calculation -

$$\hat{p} = \frac{42}{50}$$

$$\hat{p} = 0.84$$

Answer -

0.84

conclusion -

student passes the exam is 84%

MLE estimate the probability of an event using the observed sample data.

Formula

$$\hat{p} = \frac{\text{Number of defective bulbs}}{\text{Total bulbs}}$$

Given,

Total bulbs = 100

Bulbs = 8

calculation -

$$\hat{p} = \frac{8}{100}$$

$$\hat{p} = 0.08$$

Answer

The estimated probability that a bulb is defective is 0.08 (8%).

4.

MLE estimate the Probability of an event by dividing the number of successful event by the total number of observation

Formula -

$$\hat{p} = \frac{\text{Number of success}}{\text{Total Trials}}$$

Given,

60 = Total Rolls

18 = number of sixes

calculation

$$\frac{18}{60}$$

0.3

Probability of getting 6 is 0.3 (30%)

5.

Given,

$$x = 4$$

$$w = 2$$

$$\text{Actual output} = 10$$

$$\eta = 0.01$$

$$\text{Gradient} = 4$$

i) Predicted output

$$y = w \times x$$

$$y = 2 \times 4$$

$$\boxed{y = 8}$$

ii) Error

$$\text{Error} = \text{Actual} - \text{Predicted output}$$

$$= 10 - 8$$

$$\boxed{= 2}$$

$$\text{iii) new weight} = w - (\eta \times \text{Gradient})$$

$$= 2 - (0.01 \times 4)$$

$$\boxed{= 1.96}$$